



Offline LED driver NCL30051 and dimmable constant-current driver NCL30160 by ON Semiconductor



Buck-boost DC-DC LED driver LT3791 by Linear Technology

ogy to achieve a very high efficiency compared to traditional half-bridge converters.

High-power LED drivers. Early last year, Mornsun launched KC24H-R—a high-power, step-down, constant-current LED driver source. Its key features include high efficiency of up to 95 per cent, a wide input voltage range of 5.5-46V DC, low ripple and noise (<100 mV), PWM dimming and analogue dimming, continuous short-circuit protection and large capacitive loads (1000µF). KC24H-R can be used in backlights and 12V, 24V and 36V automotive lighting, landscape lighting, special lighting controls, commercial lighting, street lighting, home lighting, etc.

Compact LED drivers. Glacial-Tech has launched a series of compact LED drivers with high power efficiency—GP-LS50PH-142C, GP-LS70PH-100C, GP-LS100PH-142C and GP-LS100PH-71C. While being small with dimensions of 177×57×37 mm³, these LED drivers have a high power-conversion efficiency rate of 91 per cent ensuring low power consumption. These fully integrated AC/DC LED driver micro-modules incorporate active PFC circuitry to deliver up to 99.4W of power to the LED with a high power factor.

AC/DC constant-current driver. Texas Instruments' TPS92310 AC/DC constant-current driver with PFC

reduces costs and shrinks to fit into high-power LED retrofit bulbs, including A19, PAR30/38 and GU10. Its AC input voltage supports all common line voltages (100V, 120V, 230V, 240V and 277V). True primary-side sensing and regulation provides accurate LED current regulation, eliminating the need for optocoupler, secondary error amplifier and associated passive components. Adaptive constant on-time provides inherent PFC with no external components, which reduces the EMI signature to ensure regulatory approval.

Smart digital control technology. NXP Semiconductors' SSL21101 is the first in a new family of GreenChip LED driver ICs featuring smart digital control technology. Incorporating intelligent digital control within a compact analogue design, SSL21101 delivers better, more accurate system performance while reducing the bill of materials, providing a good solution for non-dimmable, high-performance LED applications up to 15W.

MOSFET technology. Fairchild Semiconductor has expanded its LED lighting driver solutions portfolio to include optimised low-power offerings—FLS0116, FLS3217 and FLS3247. These drivers come with integrated MOSFETs and PFC. They are optimally designed for low-power LED applications. With the addition of integrated power

MOSFET, these devices help to minimise board space and overall component count while reducing the design time. Equipped with cycle-by-cycle current-limiting and integrated protection features such as over-temperature protection and under-voltage lockout, these devices help to significantly improve the reliability of LED lamp systems.

Tips to choose appropriate LED drivers

Product designers need to know a few important things before choosing an LED driver. They should check the input/output voltage, current flowing through the LEDs, wattage, efficiency and features like dimming and PFC.

Selecting the right LED driver requires analysis of performance specifications and special features. Therefore take a closer look at the datasheet of the product.

A very important factor to check is whether the product is Underwriters Laboratories (UL), CE, Federal Communications Commission (FCC), Certified Performance Technologist (CPT), CB and IP67 certified. These certifications are a must, as they assure that the driver is a quality product well protected from water, dust and moisture. ●

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